Delayed seed germination is widespread in the temperate flowering plants of Europe, except those adapted to high disturbance

Introduction

Developmental delay is a condition in which a developmental process is arrested even thou the environment is favorable for said process to be completed (Tuljapurkar 1990; Tuljapurkar & Wiener 2000). Developmental delay often results in slower growth rates and ultimately in a reproduction delay. This is important because early development and rapid reproduction should be, in theory, associated with high individual fitness and thus favored by fecundity selection (Donohue et al. 2010; Scott & Otto 2014). Even so, developmental delay is common in many species across the tree of life (Tuljapurkar 1990; Tuljapurkar & Wiener 2000). A widespread example of developmental delay is delayed seed germination, i.e. the fact that a seed may not germinate even if exposed to favorable moisture and temperature conditions suitable for seedling establishment and survival (Scott & Otto 2014). Delayed germination can be the consequence of two factors: inherent seed dormancy and environmentally-imposed delayed germination (Lamont & Pausas 2023). Inherent seed dormancy is a state in which physicochemical properties of the seed prevent progress towards germination even if external conditions are otherwise suitable (Finch-Savage & Leubner-Metzger 2006; Baskin & Baskin 2014). Environmentally-imposed delayed germination (i.e. imposed dormancy *sensu* Lamont and Pausas (2023)) is a consequence of the external environmental cues not being suitable to trigger germination; these cues must be replaced by suitable ones for

germination to start. These external cues include signals such as specific levels and combinations of moisture, temperature, light and chemicals (Bewley et al. 2013). These two factors (inherent and imposed) have been the subject of a long terminological debate in plant ecology and physiology (Vleeshouwers et al. 1995), but both of them are mechanisms for the same ecophysiological function: delayed seed germination. What is important is to focus on the fact that, in many plant species, the set of conditions that allow germination (i.e. the seed germination niche) is narrower than the set of conditions on which the species can occur and grow. In other words, the seeds of many species are programmed to *not* germinate under many scenarios, i.e. delayed germination is widespread in the world's flora (Baskin & Baskin 2014; Lamont & Pausas 2023), even if less dormant and earlier germinating seeds would produce more offspring and of a larger size within each year's reproductive cycle (Pincheira-Donoso & Hunt 2017).

Delayed germination has traditionally been understood as a regulator of germination phenology, integrating environmental cues from long-term seasonal variation (e.g. physiological dormancy release by overwintering) with short-term and microhabitat cues (e.g. light changes in the soil depth profile) (Finch-Savage & Leubner-Metzger 2006) to ensure that seeds germinate at the right time and in the right place to maximize the probability of subsequent seedling survival and growth (Donohue et al. 2010). Following this reasoning, delayed germination is only adaptive if there are heterogeneous/different regeneration environments with different odds of seedling survival and growth, and delayed-germination cues allow to differentiate between regeneration environments (Pausas et al. 2022). If the temporal occurrence of favorable regeneration environments is unpredictable

(e.g. erratic rain events in a desert), the *bet-hedging hypothesis* proposes that delayed germination of part of a seed population can act to distribute germination events across different times, to increase the chance that at least part of the seed population will germinate during a favorable time (Venable & Brown 1988; Gremer & Venable 2014). Alternatively, if the occurrence of favorable regeneration environments occurs regularly (e.g. snowmelt in an alpine snowbed), the *best-bet hypothesis* proposes that seeds can have specific germination requirements that track predictable environmental changes, and therefore concentrate their germination in the most favorable regeneration window. To our knowledge, the term *best-bet hypothesis* was recently coined by Pausas *et al.* (2022), but its underlying logic has had a long tradition in seed ecology, for example, in the interpretation by Baskin and Baskin (1976) of the different germination strategies of winter vs. summer annuals in response to predictable intra-annual seasonality. In the winter annual example, individuals shed their mature seeds in spring, but these seeds delay their germination until autumn, so seedlings avoid predictable drought during summer (Baskin & Baskin 1976). In this particular case, delayed germination is achieved by a requirement for summer dry-hot conditions to overcome physiological seed dormancy (Baskin & Baskin 1976), and this trait is understood to be essential for the winter annual life cycle in temperate seasonal environments (Footitt et al. 2014).

To date, the macroecological patterns of delayed germination have been studied using macroclimate as a surrogate for regeneration environment favorability (Rubio de Casas et al. 2017; Carta et al. 2022; Zhang et al. 2022; Rosbakh et al. 2023). These efforts have served to test and validate many aspects of the best-bet hypothesis as formulated in biogeographical

seed ecology (Baskin & Baskin 1976). However, as pointed out recently (Pausas et al. 2022; Lamont & Pausas 2023), a needed next step is to go beyond macroclimate and focus on the various aspects of the regeneration environment that determine its suitability and its relation to delayed seed germination. The favorability of regeneration environments can be conceptualized in terms of stress and disturbance, the major drivers of plant life histories (Grime 1979). Environmental stress during regeneration has been described as the existence of a 'bad' season for germination and seedling establishment, and has usually been linked to the two major challenges of plant life, frost and drought (Jurado & Flores 2005), that have a close association with macroclimatic conditions. Plants adapted to frost- or drought-limited ecosystems should have germination traits that ensure delayed germination during the cold or dry season (Jurado & Flores 2005; Baskin & Baskin 2014). Some examples of this include alpine plants having physiologically-dormant seeds, preventing germination during autumn and winter (Fernández-Pascual et al. 2021; Rosbakh et al. 2023); or Mediterranean species having cold-cued germination, preventing germination during the summer dry season (Doussi & Thanos 2002; Carta et al. 2022). Independent from stress, disturbance has been defined as a measurable loss of biomass in a biological community (Grime 1979; White & Jentsch 2001). The relationship of disturbance with delayed germination has been particularly studied in fire-prone ecosystems (Baker et al. 2005; Dairel & Fidelis 2020). In these systems, fire can act both as a cue for dormancy release (smoke, heat) and as a creator for optimal regeneration conditions (low competition, high resource availability, low predation, low pathogen load) (Pausas et al. 2022). Therefore, fire-cued germination delay is expected to be advantageous when fire intervals are shorter than the lifespan of the dominant competitor species, and when they are sufficiently frequent and predicted to occur within the

interval between plant maturation and senescence (Keeley et al. 2011; Pausas & Keeley 2014). However, ecological theory (Macarthur 1962; MacArthur & Wilson 1967) also predicts that highly disturbed environments should select for short-lived annual plants producing many small seeds, with high capacity for dispersal in space (Venable & Brown 1988). Given the trade-off between seed dispersal in space and seed dispersal in time via delayed germination (Chen et al. 2020), we could expect that highly disturbed environments would correlate with low seed dormancy (i.e. low germination delay), as plants would escape from high disturbance by producing many cheap and highly dispersible seeds with high germinability, to maximize the chance of germinating in a place with a favorable situation along the disturbance cycle.

Here we have studied the patterns of seed germination delay in response to the underlying drivers of regeneration environment suitability: stress and disturbance. We have done so by performing a meta-analysis of primary germination records using a large dataset representing > 17,000 records of > 1,300 species from 97 botanical families of the European flowering plants. Rather than measuring delayed germination as the presence/absence of seed dormancy (Rubio de Casas et al. 2017; Zhang et al. 2022; Rosbakh et al. 2023), our meta-analysis has considered the multi-faceted responses of seed germination to various cues including scarification, cold stratification, temperature regimes and light (Carta et al. 2022). Based on the hypothesis that stress and disturbance determine the occurrence of delayed seed germination, we have tested the following predictions: (1) delayed seed germination should correlate with species adapted to high stress and/or low disturbance, while low stress and/or high disturbance should correlate with non-delayed germination; (2) the specific germination

cues driving delayed seed germination should vary depending on the species preferences for stress and disturbance following the expectations developed in **Table 1**.

Table 1: Seed germination cues. Description of the experimental treatments used as proxies of the cues, ecological function of the cue in determining germination phenology, and prediction of the relationship between the cue and the species preferences for stress and disturbance.

GERMINATION CUE	EXPERIMENTAL TREATMENT	ECOLOGICAL FUNCTION	PREDICTION
Scarification	Abrasion of the seed coat was applied to allow water imbibition; binary (yes/no)	Delay germination until seed coat is abraded by soil movement, fire, freeze-thaw, etc.	Stronger signal for high-disturbance species (e.g. germinate after soil tilling or fire)
Cold stratification	Seeds went through a period (> 4 weeks) of cold-wet stratification (< 6ºC); binary (yes/no)	Delay germination until a cold season is passed	Stronger signal for high-stress species (e.g. avoid autumn or winter germination)
Average temperature	Weighted average of the diurnal temperature cycle, weighted by the duration of each phase; in ºC	Delay germination until the proper time of the annual temperature cycle	Stronger signal for high-stress/low-disturbance species (e.g. summer germination in alpine meadows, mires)
Alternating temperature	Different temperatures were applied during the day and night, in diurnal cycles; binary (yes/no)	Delay germination unless close to soil surface, under open vegetation and/or a low water table	Stronger signal for low-stress/low-disturbance species (e.g. germinate in vegetation gaps)
Light	Seeds were exposed to light during some part of the diurnal cycle; binary (yes/no)	Delay germination unless close to soil surface	Stronger signal for low-stress/low-disturbance species (e.g. germinate in vegetation gaps)

Methods

Seed germination dataset

We obtained a curated pan-European seed germination dataset by retrieving records from *SeedArc*, the global archive of primary seed germination data (Fernández-Pascual et al. 2023). We define a record as a germination proportion of a given seed lot of a species, recorded in response to a given set of germination cues recreated in laboratory experimental conditions. To ensure data quality, we only retrieved records belonging to (1) angiosperms, to avoid the effect of gymnosperm branch lengths on subsequent phylogenetic analysis; (2) species present in the megatree of the seed plans by Smith & Brown (2018), a tree that would be used in subsequent analysis; (3) experiments conducted with seeds collected from wild populations in Europe and neighboring regions, between 30º W and 70º E and north of 30º N; (4) laboratory experiments conducted with either agar or filter paper as substrate; (5) experiments not using specialized and uncommon treatments (e.g. sterilization, nitrates, plant hormones, UV light); (6) experiments not using complex dormancy breaking cycles with warm stratification or several steps (e.g. move-along experiments), which would be not comparable with simpler and more common experiments; (7) experiments conducted with at least 5 but no more than 1,000 seeds per experimental replicate; (8) species with data available for the ecological and disturbance indicator values (see below). Furthermore, to merge heterogeneous original records, we simplified the germination cues by merging the originally-recorded treatment levels into simpler treatment levels (**Table 1**) that are routinely recorded and comparable across germination tests: (1) scarified vs. unscarified seeds (binary, 1/0); (2) cold-stratified vs. non-stratified seeds (binary, 1/0); (3) average germination

temperature (numerical, in degrees); (4) alternating vs. constant temperatures (binary, 1/0); and (5) presence/absence of light during the experiment (binary, 1/0). These cues are proxies of underlying quantitative variables that drive the physiological responses of seeds: e.g. the heat thresholds for scarification, the length and temperature of cold stratification, the cardinal germination temperatures, the amplitude of the diurnal thermal oscillations, the length of the photoperiod, etc. Further, to merge and homogenize this dataset with the stress-disturbance dataset (see below), we standardized all species names using the World Checklist of Vascular Plants V.10 (Govaerts et al. 2021).

Stress and disturbance groups

We classified species in the germination dataset as the result of the combination of two factors (stress - disturbance) x two states (high - low). Therefore, we created four stress-disturbance groups: low stress - low disturbance, high stress - low disturbance, low stress - high disturbance and high stress - high disturbance. To perform this classification, we used the species' ecological indicator values (EIVs) for temperature, moisture, disturbance frequency and disturbance severity. Although these indicators were developed for adult plants, the species-level temperature/moisture/disturbance niches of adult plants should be a reliable proxy of the conditions in which their seeds regenerate. EIVs are a numerical ordering of species according to their optima along environmental gradients. For temperature and moisture, we compared the EIVs provided by Dengler *et al.* (2023) and Tichý *et al.* (2023) and verified that (1) their values were comparable and (2) the selection of one or the other did not affect subsequent analysis. Thus, we decided to use the values by Dengler *et al.* (2023) because they covered more of the species in the germination dataset. For the

disturbance EIVs, we used the values of disturbance frequency ($\log_{10}$ of the inverse of the estimated mean interval between two consecutive disturbance events, in centuries) and disturbance severity (a proportional value, 0/1, of biomass loss in the community during a disturbance event) provided by Midolo *et al.* (2023). In this case, we decided to use the version of the EIVs calculated by Midolo *et al.* (2023) for the herb layer, since it is in the herb layer where regeneration by seed takes place. For the species in our dataset, the indicators of disturbance frequency and severity were strongly and positively correlated (Pearson's $r > 0.7$). The indicators of temperature and moisture had a moderate negative correlation ($r = -0.4$). The indicators of disturbance (frequency/severity) and stress (temperature/moisture) were weakly correlated ($r < \pm 0.4$). The four EIVs were used to classify the species into the aforementioned four groups (**Fig. 1**). We classified a species as *low stress* if its EIVs for temperature (a proxy of freezing or heat stress) and moisture (a proxy of drought or flood stress) were above 3 or below 6 (i.e. if the species avoided extreme values of both factors); and we classified a species as *high stress* if it did not meet the previous condition. Similarly, we classified a species as *low disturbance* if its EIVs for disturbance frequency and severity were below the median value of the EIV (< 1.67 frequency and 0.41 severity), i.e. if the species avoided frequent or severe disturbances; and we classified a species as *high disturbance* if it did not meet the previous condition. We acknowledge that this approach does not consider potential different effects of temperature vs. moisture stress, and of disturbance frequency vs. severity, but our research questions are interested in the overall effect of stress and disturbance across species. It must be noted that our high stress-high disturbance does not correspond to Grime's 'missing' fourth corner (Herben et al. 2018),

as our stress groups reflect the favorability of the regeneration environment, rather than productivity.

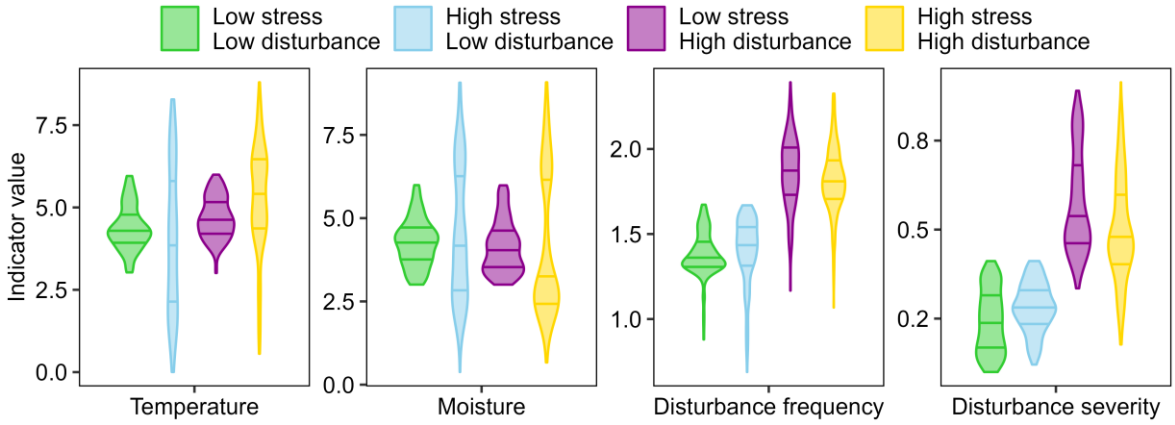

Figure 1: Stress-disturbance groups. The violin plots show the density of values for each of the four ecological indicator values (temperature, moisture, disturbance frequency and disturbance severity) in each of the four stress-disturbance groups.

Statistical analysis

We tested the relationship between species stress-disturbance groups and delayed germination by performing a meta-analysis of the germination dataset using binomial generalized mixed models with Bayesian estimation (Markov Chain Monte Carlo generalized linear mixed models, MCMCglmms) as implemented in the R package *MCMCglmm* (Hadfield 2010). See Carta *et al.* (Carta et al. 2022) for further description of the MCMCglmm methodology in the context of seed germination meta-analysis. In the models, the response variable was always the final germination proportion, i.e. the number of seeds germinated out of the number of viable seeds in the experiment. Since the conditions in which all the germination tests had been performed were generally adequate for seedling survival

and growth, we assumed that any deviation from 100% final germination represented a case of delayed germination. To test our first prediction (i.e. delayed seed germination should correlate with species adapted to high stress and low disturbance), we fitted a model for the whole dataset, with the stress and disturbance groups as predictors. To test our second prediction (i.e. the specific germination cues driving delayed seed germination should vary depending on the species preferences for stress and disturbance), we fitted separate models for subsets of the dataset representing each of the four stress-disturbance groups, with the germination cues (scarification, cold stratification, temperature, alternating temperature and light) as predictors. Predictor variables were centered and scaled so their contribution to the effect sizes could be compared. In all models, random effects included the source of the germination data (to account for between-study variation related to the research group or publication from which the data was extracted), the seed lot ID (to account for between-study variation related to differences in experimental settings, seed lot quality and storage conditions, etc.), species identity (to account for between-study variation related to differences between study species' traits) and a reconstructed phylogenetic tree for the study species (to account for the effect of the shared phylogeny). To create the phylogeny, we used the R package *U.PhyloMaker* (Jin & Qian 2023) which contains an updated mega-tree of the seed plants based on Smith & Brown (2018). We used weakly informative priors, with parameter-expanded priors for the random effects (Hadfield 2010). To take into account the potential effect of seed mass on germination responses and species ecologies, we plotted the values of seed mass (Royal Botanic Gardens, Kew 2017) across stress-disturbance groups, and repeated the models with seed mass as a covariable. The values of seed mass did not differ between stress-disturbance groups, and the inclusion of seed mass did not affect model

outputs for the study variables. Therefore, we kept seed mass out of the analyses for simplicity. Each model was run for 500,000 MCMC steps, with an initial burn-in phase of 50,000 and a thinning interval of 50 (De Villemereuil & Nakagawa 2014), resulting, on average, in 9,000 posterior distributions. From the resulting posterior distributions, we calculated mean parameter estimates and 95% highest posterior density and credible intervals (CI). We interpreted the significance of model parameters by examining CIs, considering parameters with CIs overlapping with zero as non-significant. We performed all data analyses with R version 4.3.1 (R Core Team 2023), using the R package 'tidyverse' (Wickham et al. 2019) for data processing and visualization. The original datasets, as well as R code for analysis and creation of the manuscript can be accessed at Zenodo (see Data Availability Statement).

Results

Representativeness of the germination dataset

The final germination dataset contained 17,597 experimental records obtained from 187 sources (i.e. research groups providing their primary germination data or published references from which data had been extracted). Experiments had been conducted with a total of 795,345 seeds from 5,009 seed collections originating from Europe and neighboring regions (**Appendix S1**). Seeds had been collected across the study area, with 41 countries represented in the dataset. The dataset included 1,357 species from 97 botanical families and 32 orders, representing the major clades of the angiosperms (**Appendix S1**). The dataset represented all the major European habitat types, considering both the numbers of species

(**Appendix S1**) and the number of records (**Appendix S1**). The majority of species and records came from zonal and non-anthropogenic habitat types (i.e. forests, shrublands and grasslands). Experiments had used scarified seeds in 3,393 records (19%) and cold-stratified seeds in 3,019 records (17%). The average temperatures of the experiments ranged from 0 to 40 ºC, with an average of 18 ºC. Alternating temperatures had been used in 7,377 records (42%) and light in 16,107 records (91%).

Testing the predictions

Regarding our first prediction (i.e. delayed seed germination should correlate with species adapted to high stress and low disturbance) (**Fig. 2**, **Appendix S2**), we found that, independently from the experimental conditions (i.e. across germination cues), high disturbance species had significantly higher overall germination proportions (posterior mean of the effect size = 0.416, 95% credible interval = 0.158 to 0.664, pMCMC = 0.002), i.e. species of habitats with high disturbance germinate readily across several experimental treatments and thus do not show delayed germination. High stress species, on the other hand, had final germination proportions that were not different from those of low stress species (posterior mean of the effect size = 0.060, 95% credible interval = -0.166 to 0.290, pMCMC = 0.601).

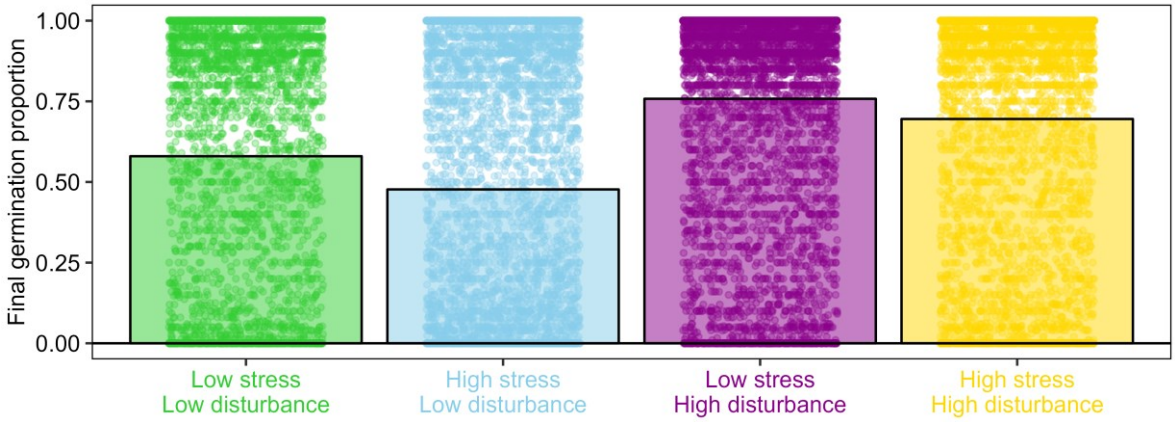

Figure 2: Seed germination of the stress-disturbance groups. The bars indicate the average final germination proportion in each of the four stress-disturbance groups, across all germination records. The dots indicate the final germination proportion in each of the original germination records.

Regarding our second prediction (i.e. the specific germination cues driving delayed seed germination should vary depending on the species preferences for stress and disturbance, **Table 1**) (**Fig. 3**, **Appendix S2**), we found that all stress-disturbance groups responded positively to all germination cues, i.e. European species germinated better after scarification, after cold stratification, at higher and alternating temperatures, and with light. However, the effect sizes of each cue varied among stress-disturbance groups, as shown by non-overlapping credible intervals. The response to scarification was weakest in the low stress-low disturbance group. The response to cold stratification was strongest in the two low disturbance groups. The response to average germination temperature was highest in the high stress-low disturbance group. The response to alternating temperature and light was strongest in the low stress-low disturbance group. In general, these results agreed with our predictions (**Table 1**) with some exceptions: (1) scarification had a strong effect in the group of species

adapted to low disturbance, but high stress; and (2) cold stratification response was more associated to low disturbance rather than high stress.

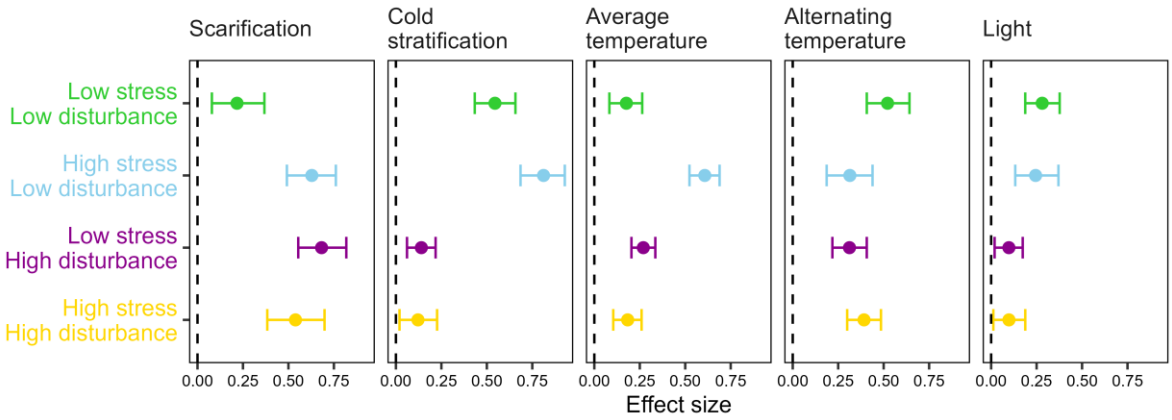

Figure 3: Germination cues in the stress-disturbance groups. Effect of germination cues (scarification, stratification, average temperature, alternating temperature and light) over the final germination proportions. Dots indicate the posterior mean of the effect size for each cue, and whiskers the 95% credible interval of the effect size. The line of zero-effect is shown: when a credible interval overlaps with the zero-effect line, the effect can be regarded as non-significant. Non-overlapping effects to the left of the zero-effect line indicate that the cue depresses seed germination, while non-overlapping effects to the left of the zero-effect line indicate that the cue promotes seed germination. In separate panels, the figure shows the effect of the germination cues in each of the four stress-disturbance models.

Random factors and phylogenetic signal

In the full model that included the entire germination dataset (i.e. the model for the first prediction), the random factor with the strongest effect was the phylogeny (mean = 3.69, CI = 2.50 - 5.00), followed by the source of the data (mean = 2.48, CI = 1.65 - 3.45), the seed

collection (mean = 2.03, CI = 1.82 - 2.25) and the species (mean = 1.33, CI = 1.01 - 1.66) (**Appendix S2**). In the models for the stress-disturbance groups, the phylogeny and data source were the first and second most important random factors, except for the model of the high stress-high disturbance group, in which data source was first and phylogeny was second (**Appendix S2**).

Discussion

Our results indicate that delayed germination, by which a seed does not germinate even in a potentially favorable environment, is widespread in the temperate flowering plants of Europe. Delayed seed germination correlates with low disturbance and is indifferent to stress. In ecological terms, this suggests that delayed germination is advantageous across the flora of Europe, except for species adapted to frequently or severely disturbed habitats. These results agree only partially with our first prediction, as we did not expect to find no differences between low and high stress groups. This suggests that delayed germination is also a feature of competitive species adapted to mild and undisturbed environments.

It must be noted that, in order to evaluate the general role of stress, we grouped together temperature and moisture stress. Potentially, this could be a reason for the lack of effect of high stress, as temperature and water stress could have different effects on delayed germination, and this could mask the overall effect of the high stress group. Nonetheless, our results indicate that low stress - low disturbance species (i.e. species avoiding at the same time temperature stress, water stress and disturbance) had lower germinability than the high disturbance groups. Species in this group are adapted to stable and mild ecosystems

(e.g. forests), where competition should be high, and factors like light may become the major drivers of plant strategies. It would seem that stable and competitive conditions also favor delayed seed germination, as in these conditions waiting for a favorable regeneration gap could become a priority, not because of seedling mortality risk, but simply to have a space to regenerate into. In any case, we need to consider that our study only included the European angiosperms. Because of the vegetation history of Europe (glaciation cycles, seasonal climate, high level of anthropogenic transformation), most local species experience stress and disturbance in one form or another. Therefore, to fully test delayed germination in relation to stress, we should consider biomes with low stress levels and floras adapted to truly stable and mild regeneration environments (e.g. tropical rainforests). Unfortunately, such floras currently have a low coverage in seed germination databases (Visscher et al. 2022; Fernández-Pascual et al. 2023).

Disturbance must be seen from the point of view of the seed. Being relatively small and defended, seeds are not so much affected by disturbance in the same way as adult plants do, i.e. by losing their aerial biomass or dying. For seeds, disturbance may represent an opening of seedling establishment opportunities, for example by reducing competition. In this sense, the importance of disturbance as a gap provider will depend on how well the frequency and severity of disturbance align with the life span of the seed and of its competitors (Keeley et al. 2011; Pausas & Keeley 2014). A systematic simulation of life history strategies using temperate grasslands as a model (Schippers et al. 2001) concluded that dormancy is superior to dispersal as a method for coping with disturbance. Our results do not support this view that, in high-disturbance environments, germination delay via seed dormancy should be

expected (Schippers et al. 2001). In our study we have used a wider gradient of disturbance considering a sample of the European flora from different habitats and life forms. Our evaluation concludes that, when looking at the continental gradient, high disturbance leads to lower delayed germination, i.e. to higher germinability. When disturbances are sufficiently frequent or severe to provide frequent or large regeneration gaps, the production of high-germinability seeds seems to be favored, possibly as a way to occupy the more readily available regeneration gaps.

Our models supported the prediction that different levels of stress and disturbance favor different responses to germination cues as drivers of delayed germination and germination phenology, but with some deviations from our initial expectations (**Table 1**). A strong response to scarification (i.e. a signal to remove an impermeable seed coat and physical dormancy Lamont & Pausas (2023)) was found in all cases except for the group of species adapted to low stress and low disturbance. This supports the role of physical dormancy as a way to detect and respond to disturbance events such as fire (Ooi et al. 2014; Pausas & Lamont 2022) but also to delay germination under unfavorable seasons such as summer for winter annuals (Van Assche et al. 2003; Van Assche & Vandelook 2010). The response to cold stratification (i.e. a signal to alleviate physiological dormancy (Finch-Savage & Leubner-Metzger 2006)) was strongest in species adapted to low disturbance levels, independently of their stress preferences. A positive response to cold stratification in species with an unsuitable season for germination (e.g. risk-avoidance of autumn-winter germination) agrees with previous findings (Jurado & Flores 2005; Rubio de Casas et al. 2017; Rosbakh et al. 2023). In addition, our results suggest that in environments without a

pronounced suboptimal season for seedling establishment, physiological dormancy can still be advantageous to delay germination, possibly as a way to avoid competition and wait longer for a regeneration gap to open. For example, delayed germination in spring would be advantageous in highly competitive deciduous forest vegetation, as this might be the only opportunity to establish in a temporarily available canopy gap (Blandino et al. 2022). Average germination temperature (i.e. an environmental signal of the annual temperature cycle (Rubio de Casas et al. 2017; Fernández-Pascual et al. 2021)) had a positive effect in all groups, i.e. species of all groups prefer relatively warm germination temperatures. Taken together with the effect of cold stratification, these results reflect that most of the studied European species are spring germinators. The effect of temperature is stronger (i.e. the preferred germination temperature is even warmer) in species from high stress-low disturbance habitats, which agrees with the warm-cued germination reported for habitats such as alpine grasslands (Fernández-Pascual et al. 2021) or mires (Fernández-Pascual 2016). Finally, alternating temperatures and light are usually interpreted as interchangeable signals of microhabitat factors such as burial depth, canopy closure or water table height (Thompson & Grime 1983; Pons 2000; Carta et al. 2017). Our results support this idea, as the effect of these two cues was strongest in species from the low stress-low disturbance group, i.e. species adapted to habitats without a marked unfavorable regeneration environment, where microhabitat detection can be most important, signaling favorable gaps such as a forest clearing after the fall of a tree. Overall, the results for the germination cues support the conclusion of the first prediction, as in general the effect of the germination cues (with the exception of scarification) was stronger in the low disturbance groups, i.e. high disturbance

species have less sensitivity to specific germination cues and thus they have less delayed germination in a wider range of conditions.

In conclusion, this study shows that delayed seed dormancy is widespread except in high disturbance situations. In all other scenarios, delayed seed germination seems to be the norm, even in stable environments without a marked unfavorable season for seed regeneration. This highlights the potential role of competition avoidance for the development of delayed germination strategies. It also suggests that there might be habitat specific factors driving the degree of delayed seed germination. In this work, we have furthered the traditional understanding that delayed seed germination is a result of unfavorable regeneration environments, by deconstructing the concept of regeneration unfavourability along axes of stress and disturbance. To concentrate on these two factors, we have left the potential effects of stress types, species traits, etc. out of focus. Future studies will need to further explore these questions, extending the analysis to non European floras, and producing curated data from macroecological experiments that carefully control for seed and adult environments, species life forms and functional traits, habitats (e.g. forests vs. shrublands vs. grasslands vs. wetlands) and separate stress factors (e.g. temperature, moisture, light, salinity). Such experiments will continue to shed light into delayed seed germination, which appears to be a widespread functional feature of the seed plants.

Data availability

The original datasets, as well as R code for analysis and creation of the manuscript can be accessed at the GitHub repository https://github.com/efernandezpascual/disturbance. Upon publication, a version of record of the repository will be deposited in Zenodo.

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1 **Supplementary**

2 **Appendix S1** Representativeness of the germination dataset.

3 **Appendix S2** Model output summaries for the MCMCglmms.